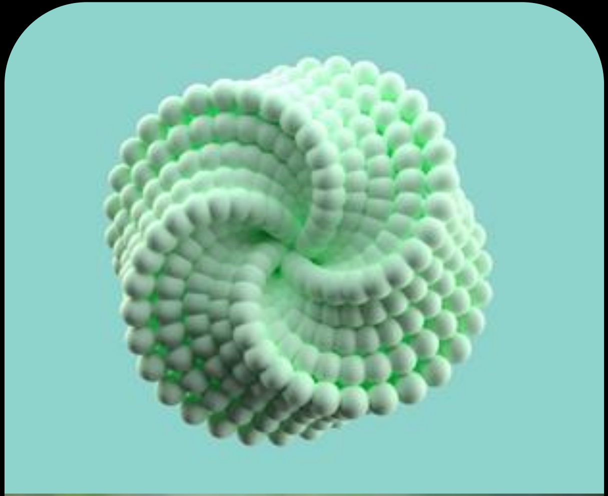

July 2nd

Superstore Sales

Customer Segments & Marketing Mix

Business Statistics & Insights
Master in Business Analytics & AI
Porto Business School
Group 4

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Agenda

INTRODUCTION

What Is This Analysis Trying to Answer?

Team 4 · Superstore Sales · Business Statistics & Insights Group Project

RESEARCH QUESTION

Which customers and marketing channels actually create business value — and how should Superstore allocate its resources and campaign spend to grow profit, not just sales?

PURPOSE OF THIS REPORT

The analysis addresses the four assignment tasks (data preparation, descriptive statistics, statistical inference, recommendations) while reflecting the analytical work expected in Finance & Marketing roles — profitability, segmentation, and channel efficiency.

DATASET AT A GLANCE

9,994

Transactions, zero rows dropped

2014–2017

Four complete calendar years

3 · 4 · 3

Segments · Regions · Categories

TECHNICAL ENVIRONMENT

Python analysis pipeline, validated in BigQuery and visualised in Looker Studio. Fully reproducible via a fixed random seed.

Business Problem

Three forces quietly eroding profitability

$2.30M in sales over four years — but only **12.5% net margin.** That gap conceals significant value destruction beneath the surface.

Indiscriminate Discounting

Discounts are applied across all segments and categories with no systematic policy.

- Orders above 20% discount lose $77–$107 on average, per order
- $567K in discount cost — 24.7% of total sales
- The single largest controllable margin leak in the business

Undifferentiated Segment Treatment

Consumer, Corporate, and Home Office are administrative labels, not behavioural profiles.

- Predefined segments show nearly identical customer behaviour
- No additional targeting signal beyond the labels themselves
- Segment-based campaigns are not backed by real differentiation

Inefficient Channel Allocation

Marketing spend does not reflect how efficiently each channel acquires customers.

- Display: $40.15 CAC for a 30x LTV:CAC return
- Email: $5.97 CAC for a 101x LTV:CAC return
- A 6.7x efficiency gap the current budget mix ignores

Data Preparation

A governance-aware pipeline: staging → intermediate → marts

Data Quality Assessment

- **0 nulls, 0 duplicates**
  Dataset confirmed clean end-to-end — no imputation or deletion required
- **0 Ship Date < Order Date violations**
  Temporal integrity confirmed across all 9,994 records
- **Profit Margin outliers flagged**
  Winsorisation applied only for downstream clustering — no records dropped
- **Discount distribution is multimodal**
  Clear spikes at 0%, 20%, 40%, 60%, 80% confirm a structured pricing policy
- **Date range 2014–2017**
  Four complete calendar years — suitable for seasonality analysis

Feature Engineering — 9 Derived Columns

Nine columns were engineered from the base dataset to enable profitability, operational and time-based analysis.

- **Profit Margin** — Profit ÷ Sales — dimensionless comparator across order sizes
- **Delivery Days** — Ship Date − Order Date — operational performance, independent of volume
- **Year / Quarter / Month** — Temporal decomposition for seasonal and trend analysis
- **Discount Category** — 4 ordered bands: 0%, 1–20%, 21–40%, >40%
- **Discount_Cost** — (Sales ÷ (1 − Discount)) × Discount — aggregates to $567K company-wide
- **State Abbreviation** — Two-letter state code for geographic visualisation

Data Preparation — Synthetic Marketing Data

Extending the dataset so channel economics can be analysed

Why: The Superstore dataset records what was sold, but not how customers were acquired or at what cost — without it, CAC, ROAS and LTV:CAC cannot be computed. Synthetic data was generated deliberately, transparently, and with academic approval, and is clearly labelled as a directional proxy, not validated fact. It only feeds the CAC / LTV:CAC analysis — never the core statistical or transactional results.

customer_acquisition

793 rows · one per customer

Acquisition Channel, CAC, Campaign label, and LTV:CAC ratio (profit ÷ CAC)

monthly_channel_spend

288 rows · 6 channels × 48 months

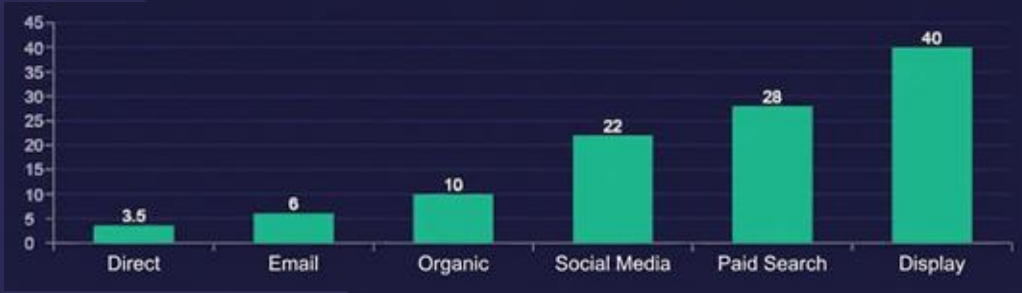
Monthly spend per channel with seasonal multipliers (higher Q4, lower Q1) for ROAS trending

enriched_with_clusters

9,994 rows · full transaction table

Original transactions enriched with Discount_Cost and customer-level Acquisition Channel & CAC

CAC Calibrated to Industry Benchmarks, by Channel



PORTFOLIO CALIBRATION

$18.51 mean CAC across the portfolio — matching the industry benchmark midpoint for multi-channel retail.

$3.32 – $40.15 range across channels, seeded with numpy (seed = 42) for full reproducibility.

CENTRAL TENDENCY · DISPERSION · DISTRIBUTION SHAPE

Central tendency (per order line)

Variable	Mean	Median	Mode
Sales	$229.86	**$54.49**	$12.96
Profit	$28.66	**$8.67**	$0.00
Profit Margin %	12.0%	**27.0%**	35.0%
Quantity	3.79	3.00	3.00
Discount	0.16	0.20	0.00

Distribution shape

Variable	Skew	Shape
Sales	12.97	Right-skewed, heavy tail
Profit	7.56	Right-skewed (leptokurtic)
Quantity	1.28	Slightly right-skewed
Discount	1.68	Multimodal (tiered levels)

Mean Sales ($229.86) is ~4× the median ($54.49); a 27% median margin collapses to a 12% mean.

COMPARATIVE ANALYSIS

The biggest region and segment aren't the most efficient

By region

Region	Sales	Profit	Margin	$/Order
West	$725K	$108K	**22%**	$67.30
East	$679K	$92K	17%	$65.33
South	$392K	$47K	16%	$56.87
Central	$501K	$40K	**−10%**	$33.79

West ≈ 2× Central per order
West earns roughly double Central's profit per order. Central is the only region whose typical order loses money (−10% margin) and it carries the heaviest discounting (24%).

By customer segment

Segment	Sales	Profit	Margin	Rev/Cust
Consumer	**$1.16M**	$134K	11%	$2,840
Corporate	$706K	$92K	12%	**$2,992**
Home Office	$430K	$60K	**14%**	$2,903

Largest doesn't equal as most valuable
Consumer supplies the most volume but the thinnest margin and lowest revenue per customer. Corporate extracts the most per customer; Home Office earns the highest margin.

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COMPARATIVE ANALYSIS

Net profit by sub-category (selected)

Sub-Category	Total Profit
LOSS-MAKERS	
Tables	−$17,725
Bookcases	−$3,473
Supplies	−$1,189
TOP PERFORMERS	
Copiers	+$55,618
Phones	+$44,516
Accessories	+$41,937

Furniture 3.9% avg margin vs Technology 15.6% · Office Supplies 13.8%. Tables & Bookcases lose money regardless of who buys them.

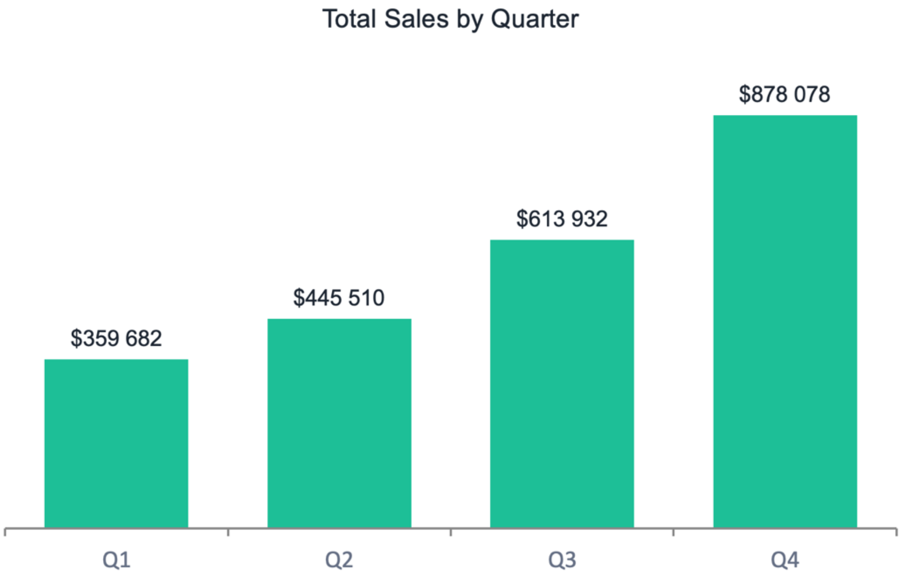
By discount level

Discount Band	Orders	Avg Qty	Avg Margin
No Discount	4,798	3.81	34.0%
Low (1–20%)	3,803	3.74	17.4%
Medium (21–40%)	460	3.78	−16.7%
High (>40%)	933	3.90	−108.9%

Avg quantity holds near 3.8 across every band (discount–quantity $\rho \approx 0$), while margin runs 34% → 17% → −17% → −109%.
Discount–margin $\rho = -0.65$: past a 20% discount the average order is loss-making.

SALES TRENDS · SEASONALITY & GROWTH

Q4 doubles Q1 but the growth is volume-led, not margin-led



Quarter	Sales	Profit	Margin
Q1	$360K	$48K	13.4%
Q2	$446K	$55K	12.4%
Q3	$614K	$72K	11.8%
Q4	**$878K**	**$111K**	12.6%

Growth is volume, not margin

November is the biggest sales month ($352K); December the peak profit month ($43K). Margin stays flat all year (11.8–13.4%) so scaling the current mix would scale the losses with it.

SYNTHESIS

Three problems the descriptive evidence forces to the surface

1

Revenue isn't converting to profit

Overall margin is only ~12.5%. The median order earns 27%, but 18.7% of orders (1,871) sell at a loss — a healthy core eroded by a concentrated pocket of loss-makers.

2

Discounting erodes margin without buying volume

Discounts show ~zero link with quantity (ρ ≈ 0) but a strong negative link with margin (ρ = −0.65). Beyond a 20% discount, the average order is loss-making.

3

Value destroyed by specific products & regions

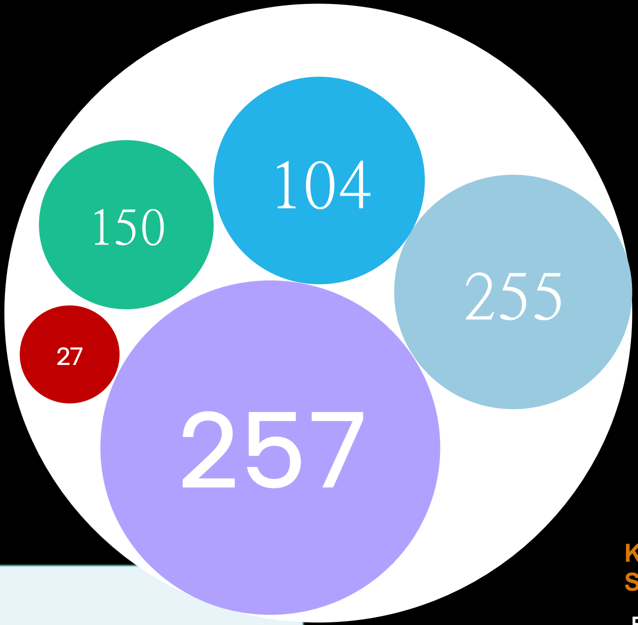
Furniture — Tables (−$17.7K) and Bookcases (−$3.5K) and the heavily-discounted Central region drag down otherwise healthy performance.

04 · CUSTOMER SEGMENTATION —RFM

RFM_Score = R + F + M ranges from 3 (worst) to 15 (best)

Market Size

- VIP/ Champions
- Loyal Customers
- Potential Loyalists
- Needs Attention
- At Risk (loss-making)



RFM Framework

R	**Days since last purchase** *Recent buyers are more likely to respond*
F	**# distinct orders placed** *Repeat orders signal loyalty*
M	**Total profit generated** *Profit-based CLV > revenue-based CLV*

```
RFM Score distribution:
count    793.00
mean       9.01
std        3.01
min        3.00
25%        7.00
50%        9.00
75%       11.00
max       15.00
Name: RFM_Score, dtype: float64

Score range: 3 to 15
```

The Core Question

Which customer segments create the most value for the Superstore?

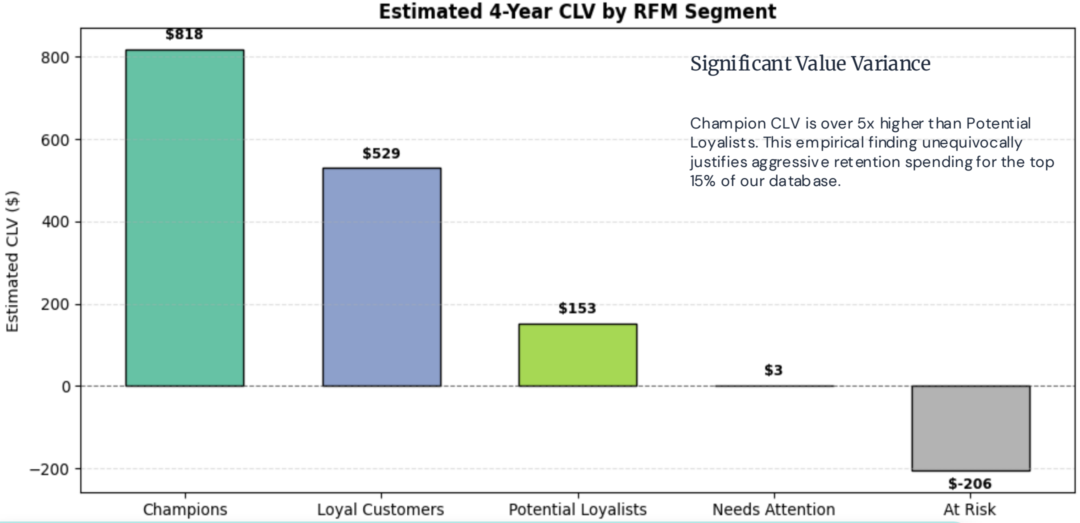
Why profit and not sales? A customer spending $10,000 at heavy discounts may generate less profit than one spending $3,000 at full price. Profit-based monetary value aligns segmentation with true commercial value.

Key Negative Finding – DB Segments

Predefined segment: (Consumer/Corporate/Home Office) split ~52/30/19% in EVERY cluster — identical to overall proportions. Segment label adds zero incremental behavioural signal. RFM clusters are strictly superior for targeting.

```
Reference date: 2017-12-31

Customers in dataset: 793
       Recency  Frequency  Monetary
count   793.00     793.00    793.00
mean    147.80       6.32    361.16
std     186.21       2.55    894.26
min       1.00       1.00  -6626.39
25%      31.00       5.00     36.61
50%      76.00       6.00    227.83
75%     184.00       8.00    560.01
max    1166.00      17.00   8981.32
```

04 · CUSTOMER SEGMENTATION — CLV BY SEGMENT



$818

LTV (Customer Lifetime Value) is an estimate of the total revenue a business can expect from a single customer throughout their relationship

AVG. CHAMPION CLV

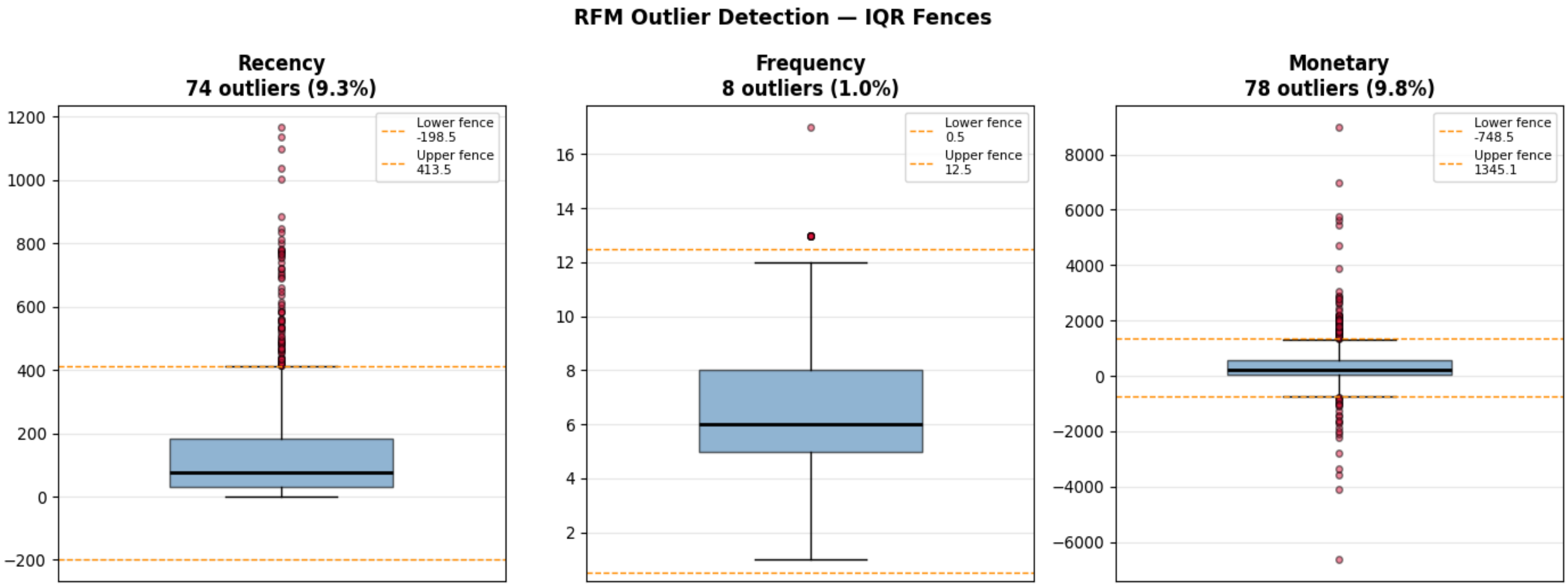
Quintile scoring (1–5 each) · Composite RFM Score = R + F + M (range 3–15)

Label	Score	N	%
Champions	13–15	104	13.1%
Loyal Customers	10–12	255	32.2%
Potential Loyalists	7–9	257	32.4%
Needs Attention	4–6	150	18.9%
At Risk	3	27	3.4%

Methodology Pipeline

1. **Build RFM table**
   793 rows (one per customer)
2. **Outlier detection (IQR)**
   Monetary & Recency fenced (~10% flagged)
3. **Winsorisation**
   Cap at fence values — 0 rows dropped
4. **StandardScaler**
   Mean=0, Std=1 — equal Euclidean weight
5. **k-selection**
   Elbow + Silhouette peak: k = 5 (score 0.348)
6. **KMeans(k=5)**
   n_init=10, seed=42, inertia=732.60

04b · K-MEANS CLUSTERING



- **Monetary** has the most severe outliers: the negative floor (−$6,626) represents customers who generated net losses; the upper outliers (>$1,345) are high-value VIPs. Together they account for ~10% of customers.
- **Recency** outliers (>414 days) are customers inactive for over a year — genuinely lapsed, but not erroneous data.
- **Frequency** outliers (>12 orders) are a small group (~1%) of very frequent buyers.

These are **real business behaviours**, not data errors. We keep them in the dataset but cap their influence on the clustering algorithm using Winsorisation.

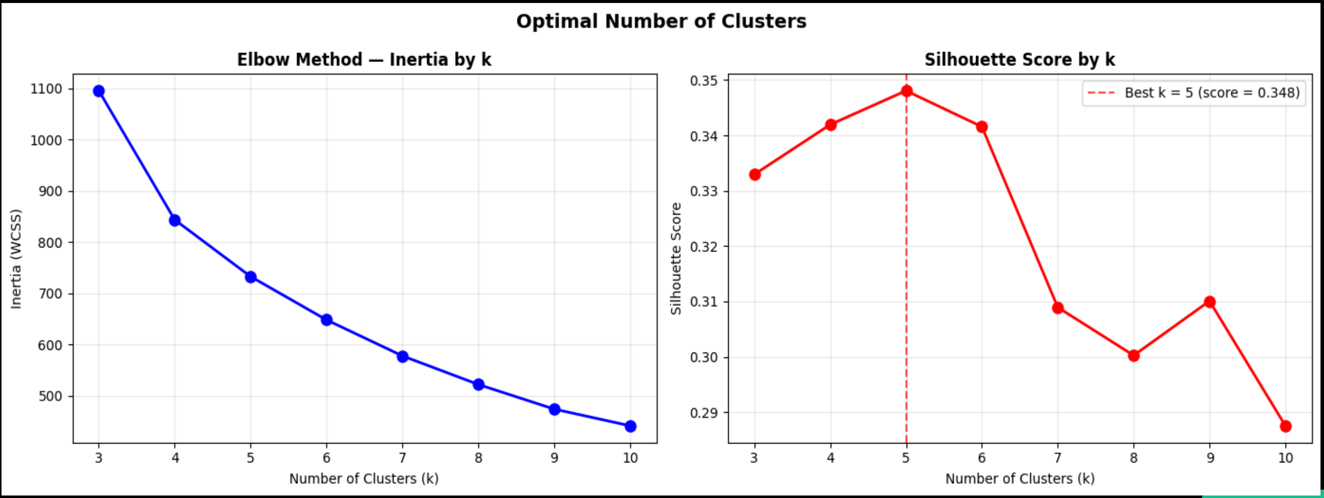
04 · CUSTOMER SEGMENTATION — RFM WINSORISATION

Winsorisation (capping values at the fence)

Instead of deleting outlier customers (which loses information), winsorisation **caps** their values at the IQR fence. A customer with $50,000 in profit becomes $8,981 (the upper fence) — they're still in the dataset, still counted, but their extreme value no longer drags the cluster centroid toward them.

Cluster Profiles

Cluster	Size	Recency	Freq.	Monetary
VIP / Champions	126	94d	7.8	$1,151
Loyal Buyers	210	64d	8.8	$319
Mid-value	246	76d	4.6	$190
Lapsed	146	367d	4.1	$173
At Risk (loss-making)	65	69d	7.0	−$460

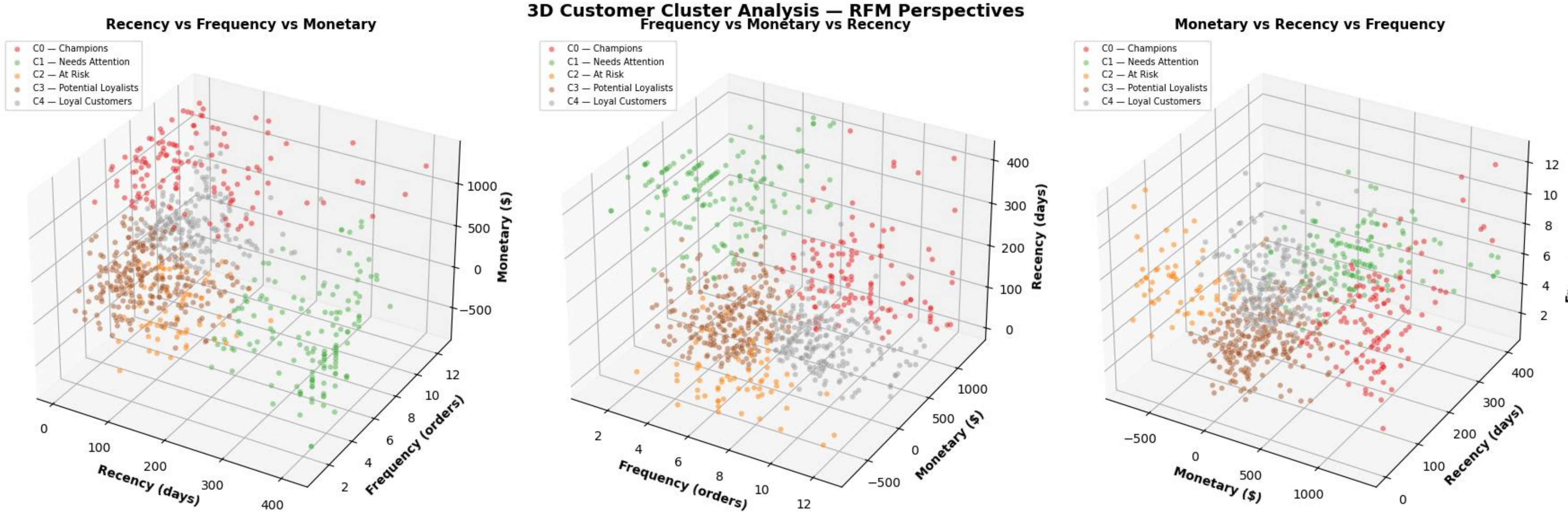



```
# Final K-Means fit with best_k
# n_init=10: run 10 random initialisations and keep t
best result
# random_state=42: ensures reproducibility
k_final = best_k

km_final = KMeans(n_clusters=k_final,
random_state=42, n_init=10)
rfm_w['Cluster'] = km_final.fit_predict(features_scaled)

print(f'Final model: k = {k_final}')
print(f'Inertia: {km_final.inertia_:.2f}')
print('\nCluster sizes:')
print(rfm_w['Cluster'].value_counts().sort_index())
```

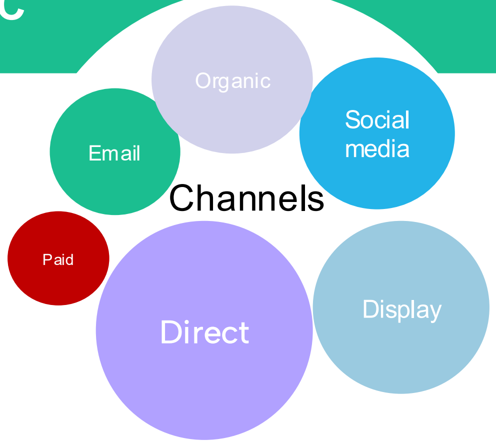
```
Final model: k = 5
Inertia: 732.60

Cluster sizes:
Cluster
0    126
1    146
2     65
3    246
4    210
Name: count, dtype: int64
```

Best k by silhouette score: 5 |
Score: 0.348

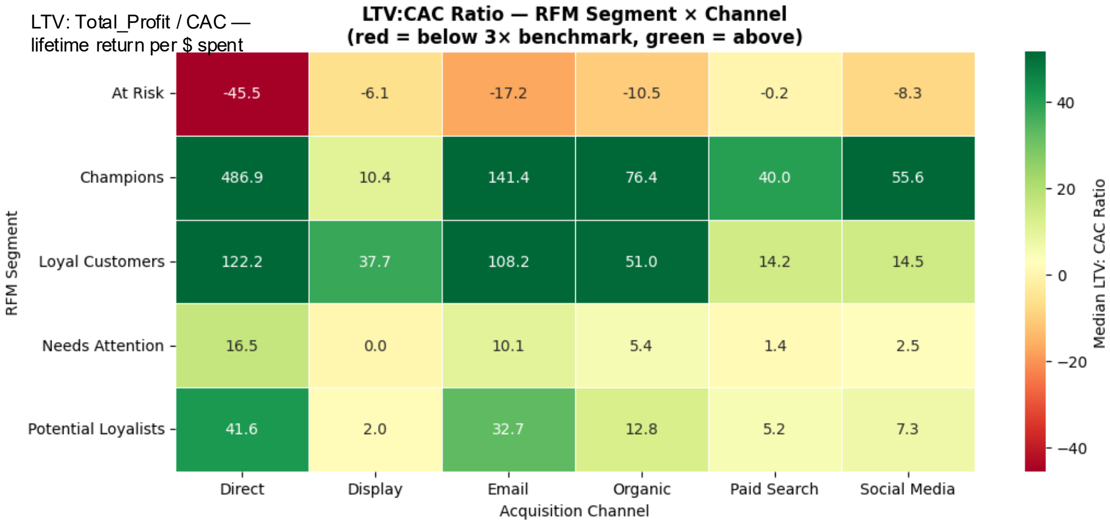
04 · CUSTOMER SEGMENTATION — Scatter plots: Frequency vs Monetary and Recency vs Monetary

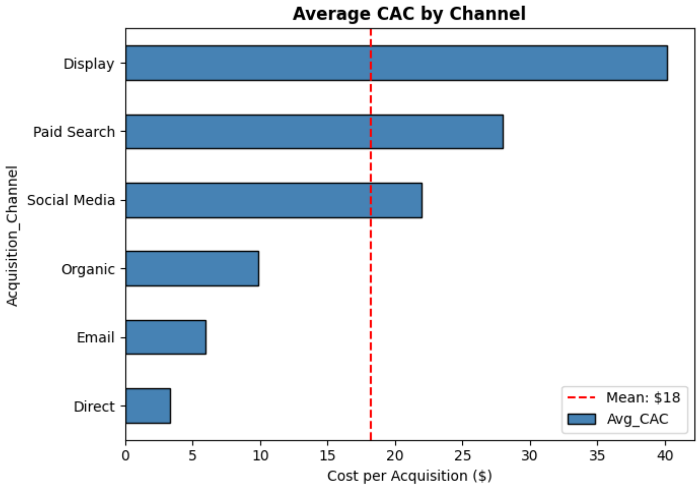


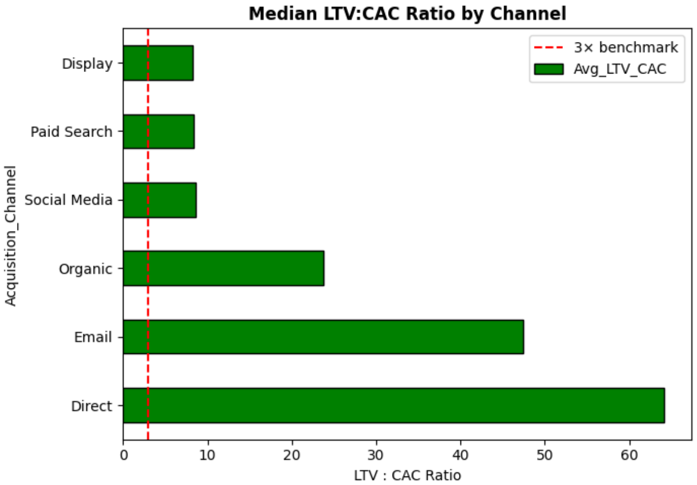
04c · MARKETING MIX & LTV & CAC



LTV: Total_Profit / CAC — lifetime return per $ spent



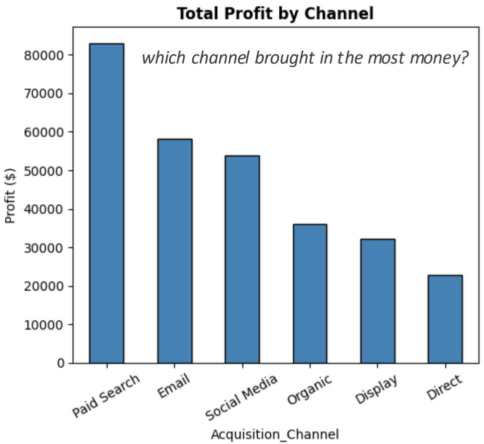


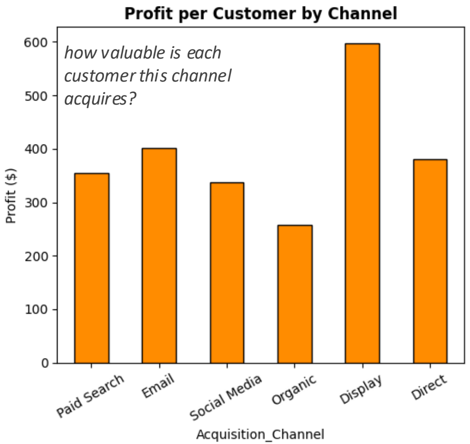


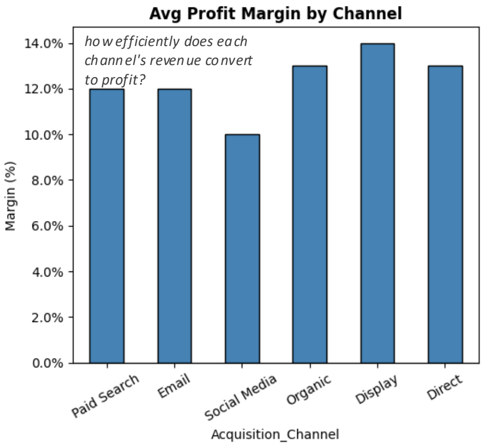
LTV (Customer Lifetime Value) is an estimate of the total revenue a business can expect from a single customer throughout their relationship

Why it matters? Profitability Benchmarking: Marketers compare LTV against Customer Acquisition Cost (CAC). A healthy business typically maintains an LTV:CAC ratio of 3:1 or higher

04d · MARKETING MIX & BUDGET ALLOCATION



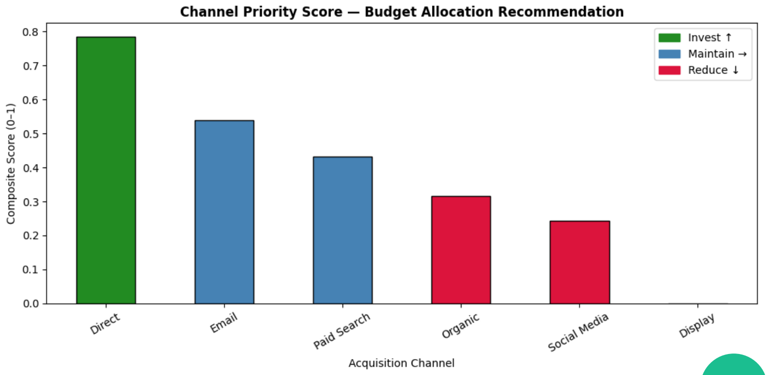




Budget Allocation Waterfall

Where should our markenting strategy next year's spend go?

Channel	Customers	Avg CAC	LTV: CAC	Action
Direct	60	$3.32	174×	↑ Invest — referral programme
Email	145	$5.97	101×	↑ Increase budget allocation
Organic	140	$9.89	39×	↑ SEO & content investment
Social Media	160	$21.97	54×	→ Monitor CAC closely
Paid Search	234	$27.98	36×	→ Optimise targeting
Display	54	$40.15	30×	↓ Reallocate 30% to Email



Annual ROAS: Direct 14× · Email 5× · Paid Search 31× · Social 12× · Display 3× · Organic 3× (synthetic data, calibrated to industry benchmarks)

05. STATISTICAL INFERENCE

Testing whether differences in sales, profit and purchasing behavior across customer groups are statistically real or just noise. Two frameworks are compared: Registration Segments (Consumer / Corporate / Home Office) and RFM Behavioral Segments (Champions, Loyal, Potential Loyalists, Needs Attention, At Risk). Significance level $\alpha = 0.05$ throughout.

WHAT WE DID - SIX STAGES OF INFERENTIAL ANALYSIS

01

Normality Check

QQ plots & D'Agostino & Pearson on Sales, Profit, Profit Margin, Discount

Result confirmed non-normal distribution,. Non-parametric tests (Kruskal-Wallis) are used as the primary inferential framework. Parametric tests (ANOVA) are included for comparative reference only.

02

Confidence Intervals

95% CIs for Sales, Profit, Margin, Discount & Delivery across both segments: mean (t-dist) + median (bootstrap)

Result indicate that the gap between mean and median CIs across segments confirms that transactions with deep discounts, distort average profitability. Discount policy is the primary issue management should address

03

Kruskal-Wallis & ANOVA

8 non-parametric + 8 parametric tests across both segmentation frameworks

- **Null Hypothesis (H_0):** All groups have the same distribution
- **Alternative (H_1):** At least one group differs significantly
- Significance threshold: $p < 0.05$ → Reject H_0

Results indicate that RFM segmentation significantly differentiates customers across all performance metrics

04

Post-Hoc Dunn's Test

Pairwise comparisons with Bonferroni correction to pinpoint which segments differ.

Result indicate that Registration Segments have limited differentiation. Only Consumer vs Home Office is statistically significant across Profit Margin & Discount.
RFM Segments have clear, actionable hierarchies.

05

Chi-Square Tests

10 tests: discount, category, shipping mode, Region & season × Registration and RFM.

Result indicate that registration segments, significant associations only for shipping mode and seasonal buying with very small practical effects.
For RFM segments showed significant associations across all tested variables, demonstrating that behavioral segmentation captures meaningful differences in customer purchasing patterns and is a more effective basis for targeted business strategies than registration-based segmentation.

06

Spearman Correlation

Discount vs Quantity and Discount vs Profit Margin, non-parametric (ρ)

Results show that discounts do not increase purchase quantity across either registration or RFM segments, with only a negligible positive effect for the "Needs Attention" segment.
Conversely, higher discounts are strongly associated with lower profit margins across all customer segments, indicating that the current discounting strategy reduces profitability without generating meaningful increases in sales volume.

Note: Segment and RFM label are customer-level attributes propagated across transactions, within-group observations are not fully independent, which may inflate apparent significance.

KEY FINDING 1

RFM Segmentation Reveals Real Value Gaps and Registration Segments Don't

Registration Segments - Not Significant

Profit (p = 0.41) and Sales (p = 0.71) show no significant difference across Consumer, Corporate, Home Office. Margin and Discount are technically significant but weak, driven mainly by Home Office's lower discount reliance.

[redacted]

RFM Segments - Significant in All 4 Metrics

Profit, Sales, Profit Margin % and Discount are all highly significant ($p \approx 0$) across Champions, Loyal, Potential Loyalists, Needs Attention, At Risk which was confirmed by both Kruskal-Wallis and ANOVA.

MEAN PROFIT: 95% CONFIDENCE INTERVAL

Champions

CI entirely above $0 - consistently profitable

At Risk

[-$71, -$17] entirely negative; losses are systematic, not occasional

DUNN'S POST-HOC: PROFIT HIERARCHY (ALL PAIRS SIGNIFICANT)

Metric	Hierarchy (High → Low)	Key Significance
Profit	Champions > Loyal > Potential > Needs Attn. > At Risk	*All pairs significant, a clean staircase*
Sales	Champions / Loyal > Potential (At Risk volatile)	*Fewer differences, volume doesn't explain value*
Margin	Tier 1: Champions/Loyal > Tier 2: Potential/Needs > Tier 3: At Risk	*Cross-tier differences highly significant ($p < 0.01$)*
Discount	At Risk > Potential/Needs > Champions/Loyal	*Mirror image of the Margin hierarchy*

Business takeaway: Retire registration-based segmentation (Consumer/Corporate/Home Office) as a marketing lever as it doesn't separate financial performance. RFM behavioural segmentation does, and should drive targeting.

KEY FINDING 2

Discounting Destroys Margin and Doesn't Even Buy Volume

CHI-SQUARE

Discount Band × RFM

$\chi^2 = 103.16, p \approx 0$

At Risk customers sit in the Very-High discount band (>40%) on ~21% of orders, nearly 3× Champions' rate (7.5%). Registration segment shows no such pattern ($p = 0.38$).

SPEARMAN ρ

Discount × Quantity

$\rho \approx 0$ (7 of 8 segments n.s.)

Discounting does not meaningfully drive order volume in almost any segment. The one weak exception (Needs Attention, $\rho = 0.106$) explains under 2% of variation.

SPEARMAN ρ

Discount × Profit Margin

$\rho = -0.60$ to -0.80

Strongly negative across every segment tested ($p < 0.001$, all 8). At Risk shows the steepest erosion ($\rho = -0.80$) — deepest discounts, weakest margins.

Seasonality differs by value tier: Champions & Loyal Customers are Q4-concentrated (~41–42% of orders); At Risk customers peak earlier, in Q2, and show a much weaker Q4 lift.

CONCLUSION FROM INFERENTIAL ANALYSIS

1

Control discounts

Discount policy is the single largest controllable driver of profitability loss. Spearman $\rho = -0.80$ between discount level and profit margin for At Risk customers, combined with a mean profit CI of [−$71, −$17], makes the case for an immediate cap on Very High discounts (>40%) for all segments.

2

Retire registration-based targeting

Consumer, Corporate and Home Office are statistically indistinguishable in Profit and Sales. Redirect campaign budget toward RFM-based targeting instead.

3

Restructure the campaign calendar

Time premium offers to Champions/Loyal in Q4 (their natural peak); shift At Risk to early-year (Q1–Q2) reactivation before disengagement compounds.

Conclusion
& Recommendations

06 · RECOMMENDATIONS

Three actions grounded in data — what the CEO should implement tomorrow

01

Cap All Discounts at 20%

Eliminate on Tables, Bookcases, Supplies

- Spearman ρ=−0.645 confirms discount drives margin erosion
- 1,137 orders >20% → avg loss $77–$107/order
- Tables alone destroys $17.7K profit on $207K sales

$50–70K/year margin recovery

KPI: Neg-margin order share < 5% in 2 quarters

02

Reallocate 30% of Display Budget

Shift to Email and Direct referral programme

- Display: $40.15 CAC · 30× LTV:CAC
- Email: $5.97 CAC · 101× LTV:CAC (6.7× better)
- Customer quality comparable across channels

Blended CAC: $18.51 → below $15

KPI: Monthly CAC by channel

03

Retention Programme — 177 Customers

Target Needs Attention (150) + At Risk (27)

- Needs Attention: 248d lapsed, 3.8 prior orders
- Re-engagement cost << new-customer CAC ($18.31)
- 20% reactivation → $4,500 incremental profit

20% reactivation rate in 90 days

KPI: 90-day first-order rate post-campaign

06 · CONCLUSIONS

Key Conclusions

- 12.5% margin conceals stark sub-category bifurcation (Labels +44% vs Tables −9%)
- Predefined segments add zero incremental targeting signal vs RFM clusters
- Discounting >20% is systematic, policy-driven value destruction
- Champions (13% of customers) generate ~37% of profit — Pareto concentration
- Email & Direct: 3–6× better LTV:CAC than Display channel
- BigQuery + Looker Studio infrastructure enables live ongoing monitoring

Estimated impact: $50–70K annual margin recovery · Blended CAC below $15 · $4,500+ incremental profit from reactivation

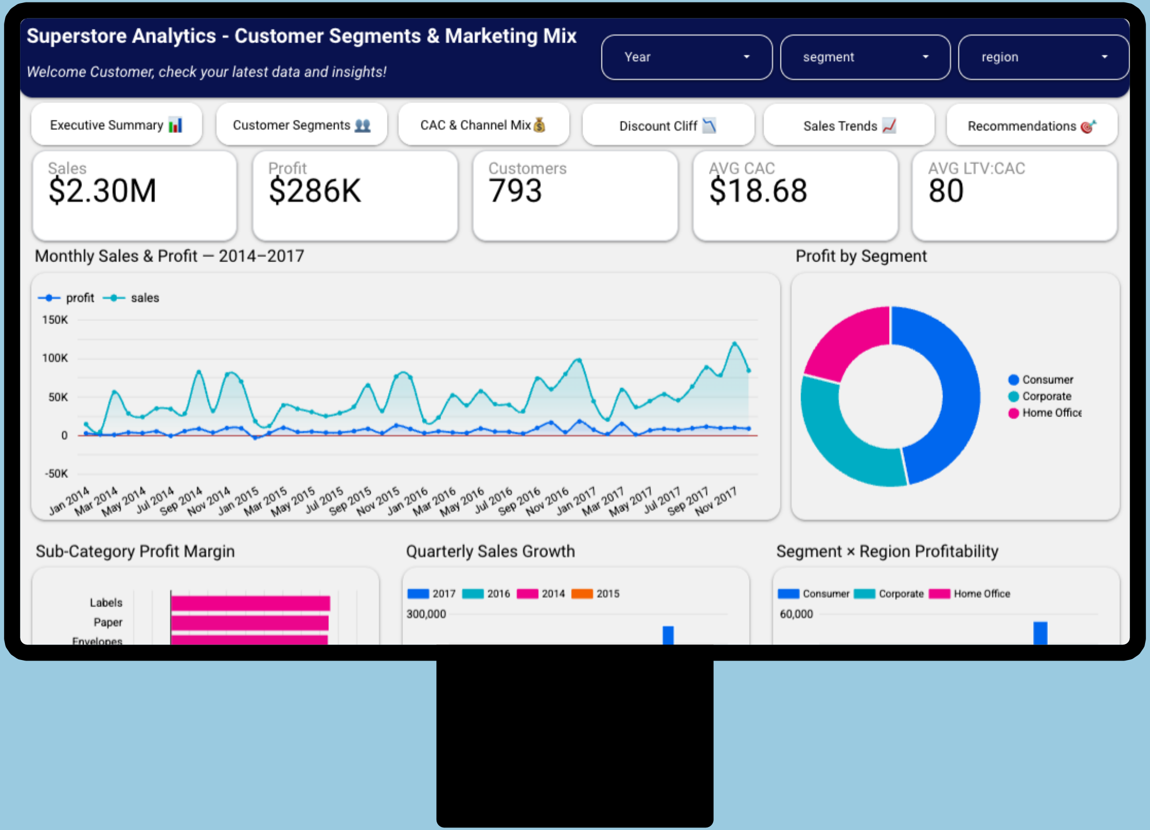
Limitations

- CAC is synthetic (directional, not validated)
- CLV proxy understates recent customers
- Dataset is 2014–2017; market norms have shifted
- K-Means assumes spherical, equal clusters

Future Work

- Prophet forecast for 2018 by category
- BG/NBD probabilistic CLV model
- Random Forest order-level profit prediction
- A/B test: Furniture discount cap
- Real marketing attribution data integration

Dashboard and KPIs for Management ➡️



[Dashboard: https://datastudio.google.com/s/hrlTCDBNtw8](https://datastudio.google.com/s/hrlTCDBNtw8)

[Repo: https://github.com/ivanya93/superstore_segmentation](https://github.com/ivanya93/superstore_segmentation)